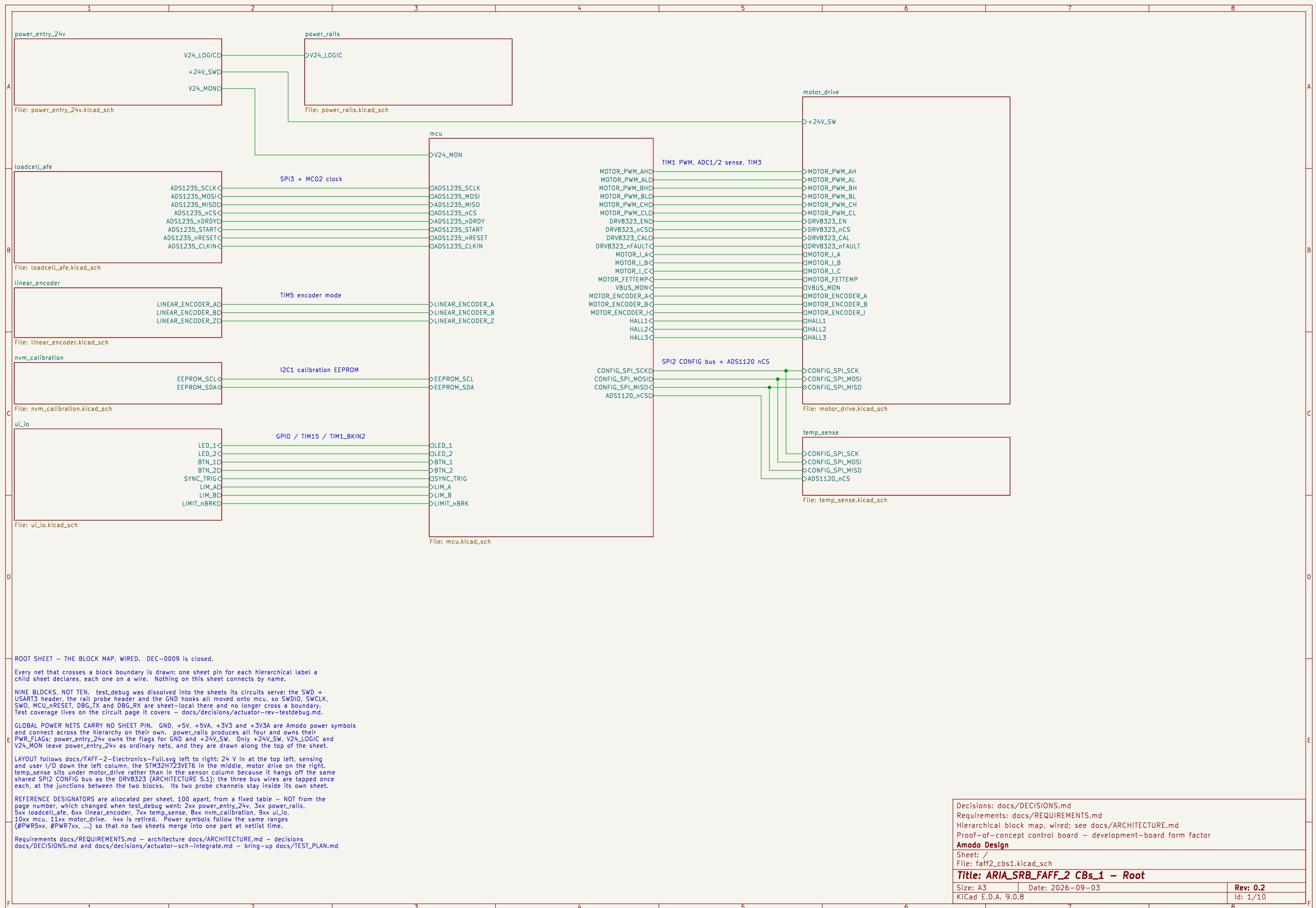
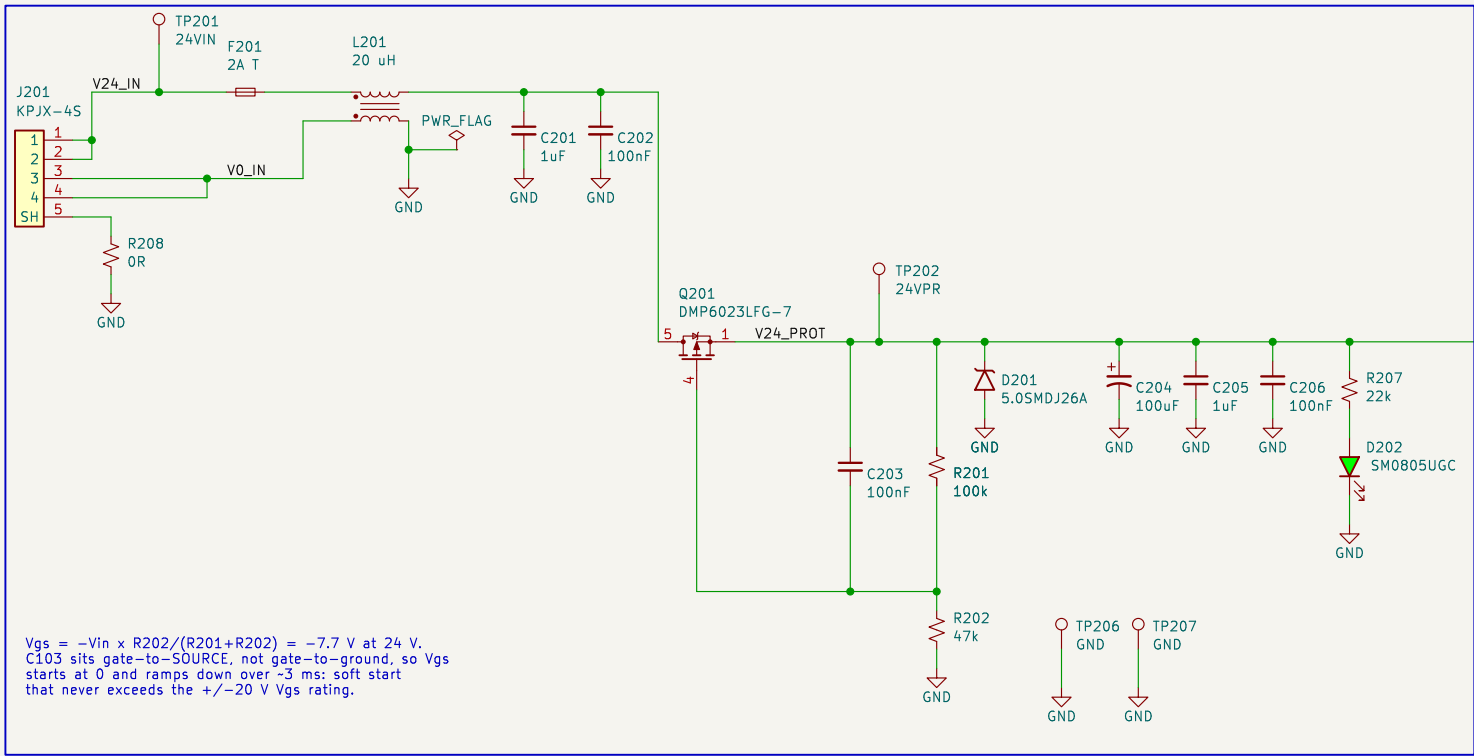
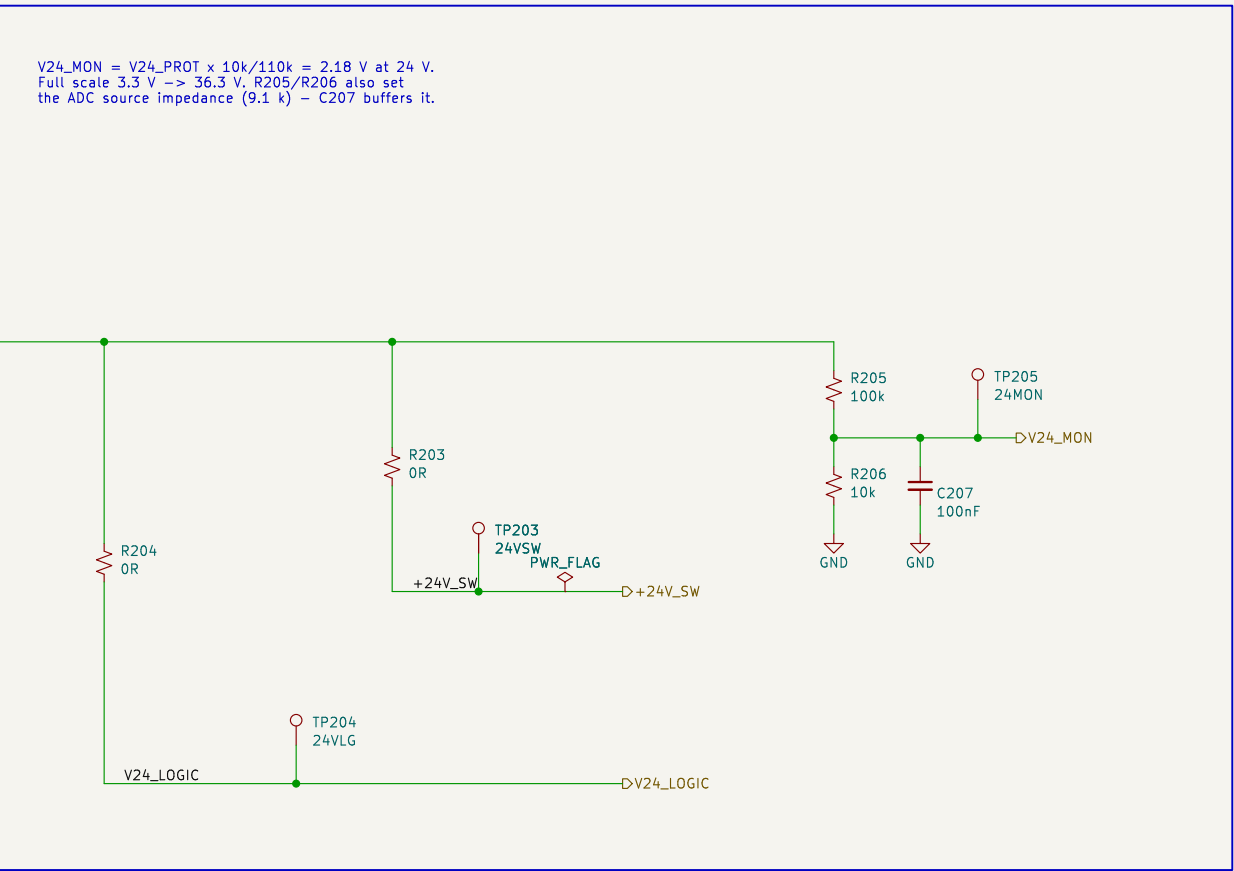




A. 24 V INPUT, FUSING, CM FILTER AND REVERSE-POLARITY PROTECTION



B. BRANCH ISOLATION LINKS / CURRENT BREAKS, RAIL MONITOR AND TEST POINTS



BLOCK: power_entry_24v - 24 V DC input and protection (REQ-EL-01..04)

CHAIN J201 KPJX-4S (contacts doubled: 1,2 = +24 V, 3,4 = 0 V, 5 = shell)
-> F201 2 A time-lag (25 W peak = 1.04 A at 24 V; 2 A T holds 1.5 A)
-> L201 common-mode choke, both lines; board GND is defined on its load side
-> Q201 P-channel reverse-polarity series switch (drain = supply, source = load, so the body diode blocks a reversed supply and nothing downstream conducts)
-> D201 TVS + C204..C206 bulk/HF -> V24_PROT, the protected 24 V bus.

BRANCHES (docs/TEST_PLAN.md 3.3 - each an isolation link AND a current break)
R203 -> +24V_SW : motor_drive only. Open R203 and the whole board can be exercised with no possibility of the actuator moving.
R204 -> V24_LOGIC: power_rails only. Independent of R203.
One OR link per branch serves as both the isolation link and the current break; a second series part would add joints and no capability.

GROUND One GND net board-wide. AGND / DGND / PGND in the block stub notes are layout partitions of it, not separate nets - see the decisions file.
This sheet owns the single GND PWR_FLAG; no other sheet may add one.

INTERFACES exported (hierarchical labels, wired to root sheet pins - DEC-0009)
OUT +24V_SW -> motor_drive OUT V24_LOGIC -> power_rails
OUT V24_MON -> mcu PC1 / ADC1_INP11
Global power nets used here: GND.

Reasoning, part sizing and residual ERC: docs/decisions/actuator-sch-power.md

Design reasoning: docs/decisions/actuator-sch-power.md
Requirements: REQ-EL-01..04 - docs/REQUIREMENTS.md
Independent motor / logic branch links (TEST_PLAN 3.3)
24 V in on KPJX-4S: fuse, CM choke, reverse polarity, TVS

Amodo Design

Sheet: /power_entry_24v/
File: power_entry_24v.kicad_sch

Title: CBs_1 - Power Entry 24 V

Size: A3	Date: 2026-09-02	Rev: 0.2
KiCad E.D.A. 9.0.8		Id: 2/10

A. PRE-REGULATOR V24_LOGIC -> 6.0 V (LMR33630, 400 kHz)

6.0 V, not 5.0 V: it is the ADPL42005's own characterised VIN = VOUT + 1 V point, so the LDOs meet their noise and PSRR numbers on +5VA, which carries REQ-FF-04. R303/R304 are sized so the worst-case floor clears 6.0 V, not just the nominal: 6.009 / 6.110 / 6.212 V. Was 5.5 V while the TPS7A20's 6.0 V input maximum capped it.

TP309 GND TP310 GND

B. POST-REGULATED 5 V RAILS (ADPL42005: +5V digital, +5VA analog)

EN is tied to each LDO input – the datasheet's own automatic-startup connection – so the 5 V rails follow the pre-regulator with no sequencing logic. SENSE ties to VOUT at the regulator, not past the OR link, so lifting the link cannot open the loop. PG is unused: no-connect, not tied to RAIL_PGOOD. +5VA feeds ADS1235 / ADS1120 AVDD and the AFE excitation chain; +5V feeds the IKP11 read head.

B. POST-REGULATED 5 V RAILS (ADPL42005: +5V digital, +5VA analog)

EN is tied to each LDO input – the datasheet's own automatic-startup connection – so the 5 V rails follow the pre-regulator with no sequencing logic. SENSE ties to VOUT at the regulator, not past the OR link, so lifting the link cannot open the loop. PG is unused; no-connect, not tied to RAIL_PGGOOD. +5VA feeds ADS1235 / ADS1120 AVDD and the AFE excitation chain; +5V feeds the IKP11 read head.

RAIL PROBE HEADER (TEST_PLAN 4)

One header to check every logic rail with a DMM during bring-up steps 2 and 3 of TEST_PLAN 5. It only observes; the rails, their isolation links and their current breaks are the rest of this sheet.

Pin	Signal
1	+5V
2	+3V3
3	+3V3A
4	GND
5	GND
6	GND

+24V IS DELIBERATELY NOT ON THIS HEADER. 24 V next to 3.3 V on a 2.54 mm strip is one slip away from destroying the board; power_entry_24v carries the +24V_SW test point and the current break that TEST_PLAN 4 asks for.

Moved here from mcu in review round 4. Its old note named this sheet as the alternative home; the captain chose it.

FB301 is the +3V3A isolation link and current break: lift it and the ADC digital supplies are cut loose.

POWER SOURCE DECLARATIONS

One PWR_FLAG per rail produced here, at the regulator that makes it, on the load side of that rail's OR link. No other sheet may add one for these nets.

The schematic shows the following components and connections:

- PH1-06-UA J301**: Header connector with pins 1 through 6.
- R313 100k**: Resistor connected between +3V3 and RAIL_PG0OD.
- C323 10uF**: Capacitor connected between +3V3 and GND.
- C324 100nF**: Capacitor connected between +3V3 and GND.
- R314 330R**: Resistor connected between +3V3 and D303.
- D303 SM0805URC**: Diode connected between +3V3 and RAIL_PG0OD.
- TP307 +3V3A**: Test point connected to +3V3A.
- TP308 PG0OD**: Test point connected to RAIL_PG0OD.
- TP311 GND** and **TP312 GND**: Ground test points.
- PWR_FLAG**: Flag indicating the power source declaration.
- FB301**: Isolation link and current break for +3V3A.

BLOCK: power_rails - rail set and topology: answers OQ-02 and OQ-07

RAILS +6V0 LMR33630 buck from V24_LOGIC. Pre-regulator only, local to this sheet.

+5V ADPL42005 LDO from +5V5. IKP11 read head (<65 mA plus RS-422 drive) and spare.

+5VA ADPL42005 LDO from +5V5. ADS1235 / ADS1120 AVDD and the AFE 5 V excitation and reference chain, which lives in loadcellLife - take it from +5VA, not +5V.

+3V3 LMR33630 buck from V24_LOGIC. STM32, USB3320, DRV8323 logic, QSPI, EEPROM, RS-422 receiver, rotary encoder.

+3V3A ferrite-isolated from +3V3 for the ADC digital supplies.

WHY A PRE-REGULATOR The analog 5 V must be regulated separately from the digital 5 V or the read head switching current lands on the ADC supply, and an LDO needs headroom above 5 V to do it. One buck at 6.0 V feeding two LDOs costs one converter and post-regulates both 5 V rails. 400 kHz on both bucks: 3.3 V at 1.4 MHz would need a ~98 ns on-time, too close to the minimum.

PER RAIL (docs/TEST_PLAN.md 3.1) one OR link that is BOTH the isolation link and the current-measurement break, a test point downstream of it, on a dual pad so a scope ground clips beside the probe. No test point sits on a feedback node:

R305 +5V5, R306 +5V, R307 +5VA, R312 +3V3, FB301 +3V3A, R305 +5V5, R306 +5V, R307 +5VA, R312 +3V3, FB301 +3V3A.

Per-CONSUMER links are not here: every block takes the same global rail net, so per-block breaks would need per-block rail names and would break the frozen block contracts. A block that wants one fits a OR at its own supply entry, inside its sheet.

SEQUENCING None required. The shared EN divider (16.2 V rising) holds both converters off until the 24 V input is real: the LDOs follow +5V5 and +3V3A follows +3V3.

INTERFACES IN V24_LOGIC <- power_entry_24v (owns its PWR_FLAG). RAIL_PG00D is open drain, the wired-AND of all four regulators through R315 / R316 / R317 / R318. SHEET-LOCAL: LED D303, TP308 are its only consumers until Q0-07 allocates an MCU GPIO. Global power nets produced here: +5V, +5VA, +3V3, +3V3A. GND is board-wide, one net: power_entry_24v owns its PWR_FLAG.

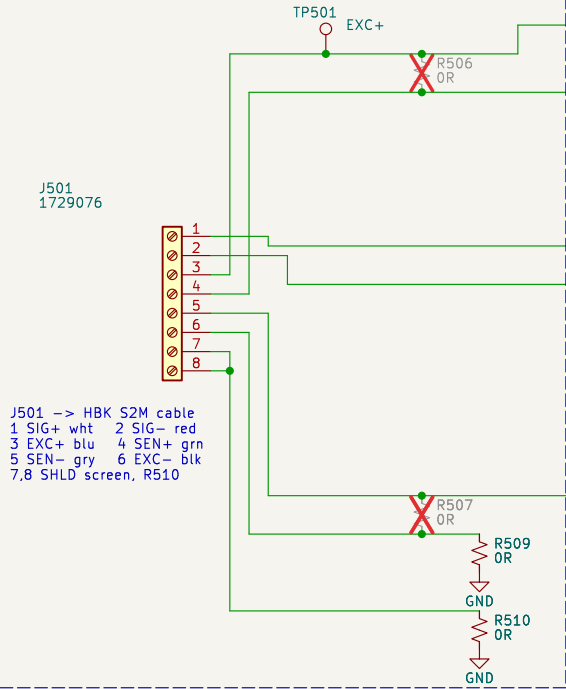
Reasoning, part sizing and residual EPC:

docs/decisions/actuators/sch-power.md

Reasoning, part sizing and residual ERC:
docs/decisions/actuators-sch-power.md

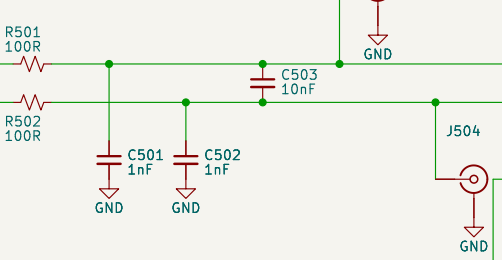
Design reasoning: docs/decisions/actuator-sch-power.md	
Rail set and topology answer QQ-02 / QQ-07	
Per-rail isolation link / current break and test point	
24 V -> 5V5 pre-reg -> +5V and +5VA; 24 V -> +3V3 -> +3V3A	
Amodo Design	
Sheet: /power_rails/	
File: power_rails.kicad_sch	
Title: CBs_1 – Power Rails	
Size: A3	Date: 2026-09-02
KiCad E.D.A. 9.0.8	Rev: 0.2
	Id: 3/10

LOAD CELL INTERFACE - 4- OR 6-WIRE

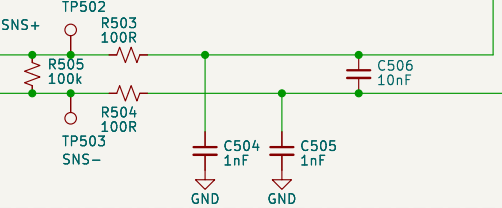


J501 -> HBK S2M cable
1 SIG+ wht 2 SIG- red
3 EXC+ blu 4 SEN+ grn
5 SEN- gry 6 EXC- blk
7,8 SHLD screen, R510

SIGNAL INPUT FILTER

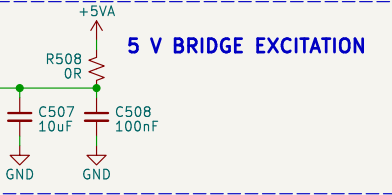


REFERENCE (SENSE) FILTER - RATIOMETRIC



REFERENCE - U501 RECEIVES IT, IT DOES NOT PROVIDE IT
Review point: is the intention that U501 outputs the reference?
No. The ADS1235 has no internal reference and no reference output pin: REFPO (32) and REFNO (31) are reference INPUTS (SBAS824 pin table; feature list: Two Reference Inputs).
What U501 measures against is the load cell's OWN excitation, sensed at the bridge: J501-4 SENSE+ / J501-5 SENSE- reach REFPO/REFNO through the R503/R504 + C504/C505/C506 filter, so VREF is the volt drop across the bridge, not a fixed voltage.
Excitation: +5VA -> R508 -> J501-3 EXC+, return J501-6 EXC- -> R509 -> GND. Signal and reference share that one source, so drift and noise divide out - ratiometric, and why +5VA need not be a precision reference (SBAS824 8.3.4.2, DEC-A5).
R505 (100k) collapses VREF with the cell unplugged: REFL_ALM turns a disconnected cable into a reported fault.

5 V BRIDGE EXCITATION



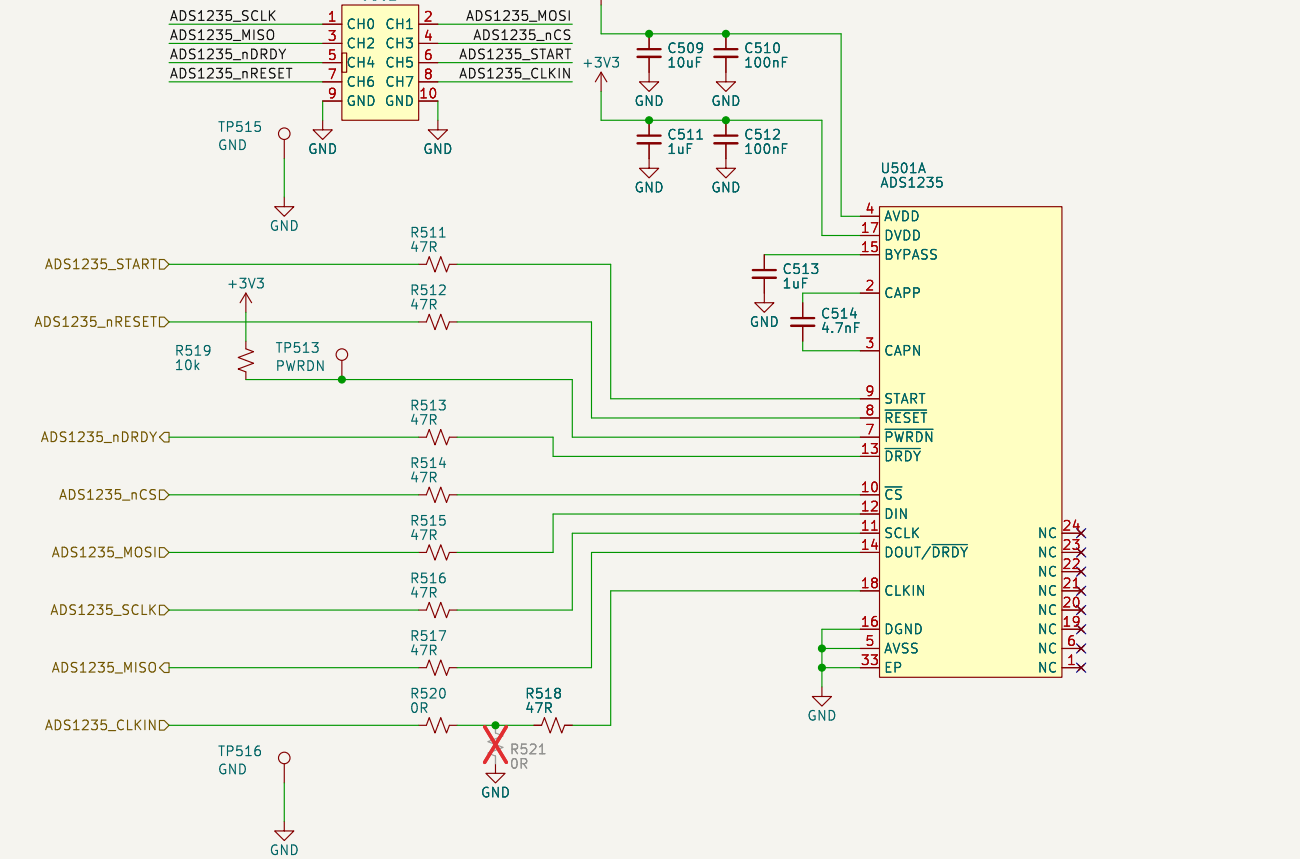
J501 PIN-OUT - HBK S2M SIX-WIRE CABLE (data sheet B03594, 2025-02-28)
1 SIG+ (white) 3 EXC+ (blue) 5 SENSE- (gray)
2 SIG- (red) 4 SENSE+ (green) 6 EXC- (black)
7 SHLD, 8 SHLD - cable screen / drain, tied to GND through R510
Bridge 350 ohm nominal, 2 mV/V, 5 V excitation -> +/-10 mV FS, -14.5 mA.
Output is positive in compression. The same terminal block accepts a resistive bridge simulator for bring-up step 6 of TEST_PLAN.md.

DEFAULT BUILD IS 6-WIRE (DEC-0014): R506 and R507 are DNP.
Fit R506 + R507 for a 4-wire cell. That ties SENSE+/- to EXC+/- at the connector, so the reference stops tracking the lead drop and the measurement loses Kelvin sensing - it is the lower-accuracy option.

OPERATING POINT (prelim calc Force sheet: ADS1235 SBAS824)
Gain 128, 4800 SPS, Sinc2, VREF = bridge excitation (ratiometric)
FSR = +/-VREF/128 = +/-39.06 mV (78.125 mV span) vs +/-10 mV signal
Input-referred noise 0.29 uVrms / 2.4 uVpp (data sheet table 1)
= 12.0 mNpp (V50N) / 2.4 mNpp (V10N) vs REQ-FF-04 20 / 5 mNpp
Ratiometric operation cancels excitation drift and noise, so the 5 V rail does not need to be a precision reference.

FILTERS - the ADS1235 data sheet figure 84 reference design.
Signal and reference filters are deliberately identical so their corner frequencies match: 100R series per leg, 1 nF COG common mode to AVSS, 10 nF COG differential. The differential capacitor is 10x the common-mode capacitors, so common-mode capacitor mismatch converts little common-mode noise into differential noise.
Differential corner ~76 kHz, common mode ~1.6 MHz; the ADC PGA antialias filter (C514, 4.7 nF COG) is at 60 kHz.
R505 (100k) across the sense pair is the data sheet reference monitor resistor: it collapses VREF if the cable is unplugged so REFL_ALM asserts. It sits on the bridge side of R503/R504 so its 50 uA does not develop an error across the reference filter.

ADS1235 SUPPLIES, SPI3 AND CONTROL

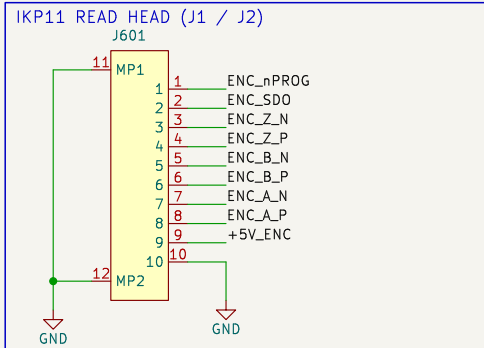


ADS1235 PIN USE
AIN4 / AIN5 = SIG+ / SIG- (bridge output, PGA gain 128)
REFPO / REFNO = SENSE+ / SENSE-, external reference, ratiometric (data sheet 8.3.4.2: "for 6-wire strain-gauge bridge applications that use excitation-sense connections, connect the excitation sense lines to the reference input pins")
AIN0, AIN3 unused - biased to AVDD through 10k (R522..R525) per data sheet 9.1.3. Resistors rather than a hard tie keep the pins safe if firmware ever configures them as GPIO or AC-excitation outputs, which are referenced to AVDD/AVSS.
PWRDN tied high through R519; reset by command or by ADS1235_nRESET. CAPP/CAPN 4.7 nF COG and BYPASS 1 uF to DGND are both mandatory. EP (footprint pad 33) to AVSS.
AVDD = +5VA (also the excitation rail), DVDD = +3V3 so the serial interface sits at STM32 logic level.

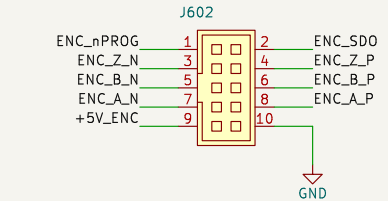
TEST AND ISOLATION PROVISIONS (TEST_PLAN.md section 4)
TP501 excitation+ TP502/TP503 sense+/- TP504/TP505 ADC reference J503/J504 U.FL on the ADC-side SIG+/SIG- nodes - the noise-floor measurement that decides REQ-FF-04. Placed symmetrically so the differential pair stays balanced (TEST_PLAN section 2.3).
J502 logic-analyser header (Amodo keyed 0.1in socket, 8510-4500PL) carries the whole digital interface on one plug: CH0..CH7 = SCLK, MOSI, MISO, nCS, nDRDY, START, nRESET, CLKIN; GND on 9 and 10. Tapped MCU-side of R511..R518, so it still sees the bus with a link lifted. TP513 (PWRDN) and TP515/TP516 GND hooks stay.
R508 excitation feed link and R509 excitation return link: open either to isolate the bridge, or replace one with a meter to measure excitation current (isolation link and current break in one part - see docs/decisions/actuator-sch-afe.md).
R510 ties the cable screen to GND at the connector; lift it to test a floating screen.
R511..R518 (47R) are the ADS1235 data sheet series terminators AND the SPI3 isolation links: remove one to take the ADC off the bus.

CLOCK - ADS1235_CLKIN COMES FROM STM32 MC02 (REQ-AR-09)
R520 fitted = MC02 drives CLKIN. R521 (DNP) instead grounds CLKIN, selecting the ADC internal 7.3728 MHz oscillator - a bring-up fallback if the MC02 clock tree is not ready. Fit one, never both.
FOR THE MCU / FIRMWARE TASK: the .ioc leaves MC02PinFreq_Value at the CubeMX default 64 MHz, outside the ADS1235 1..8 MHz CLKIN range. The clock tree must deliver 7.3728 MHz nominal on PC9.

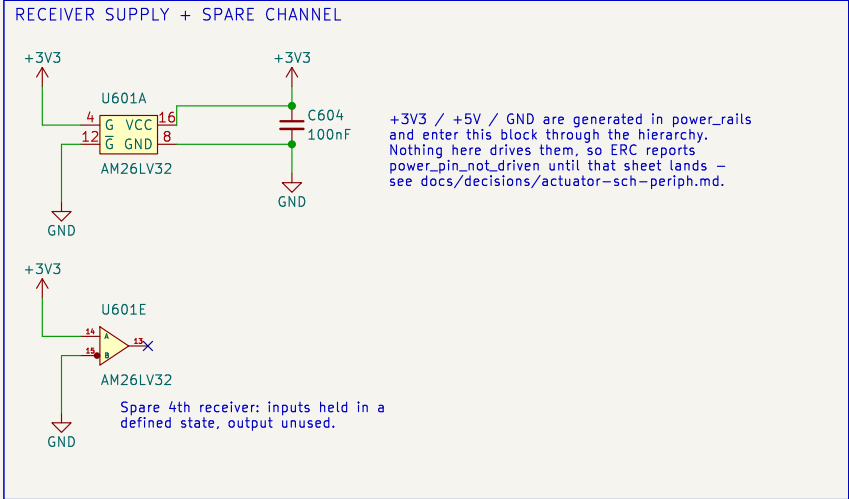
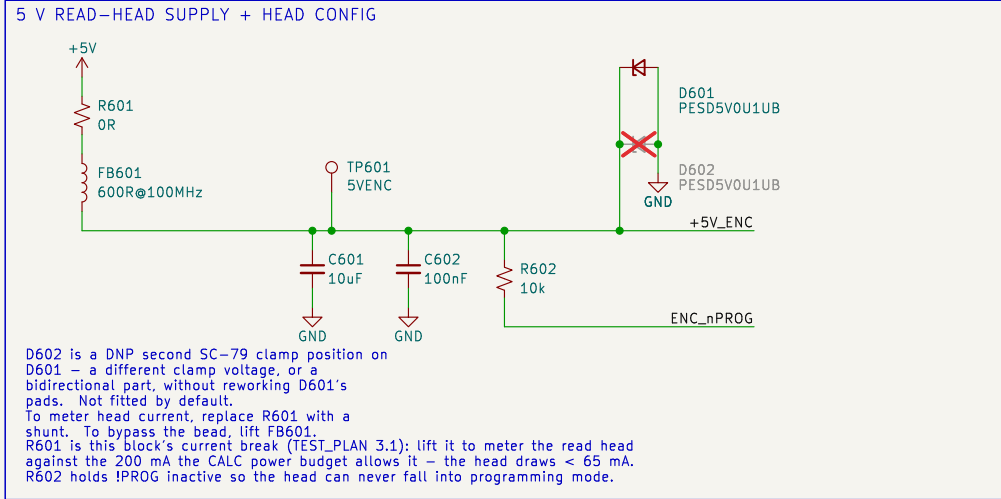
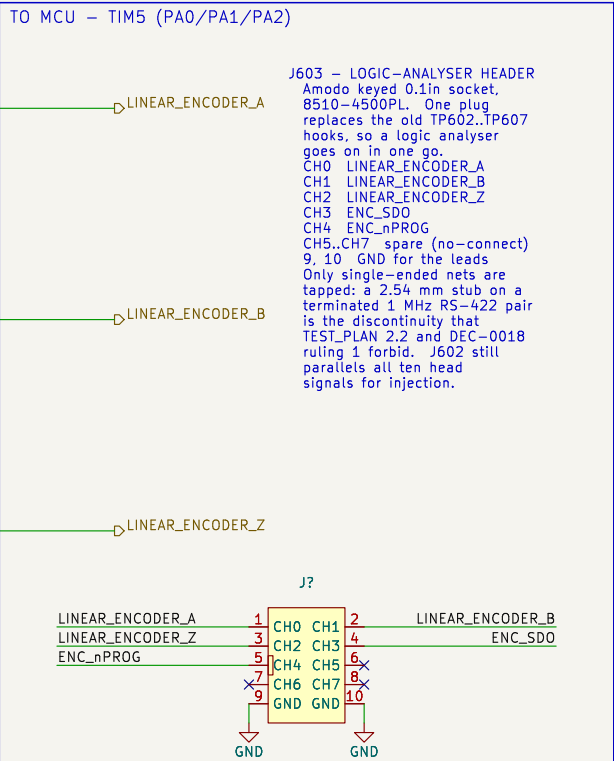
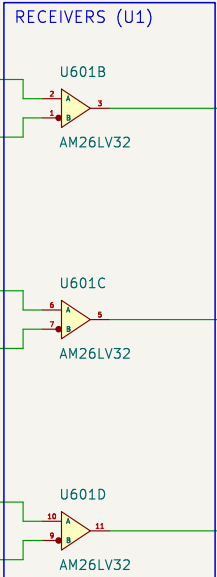
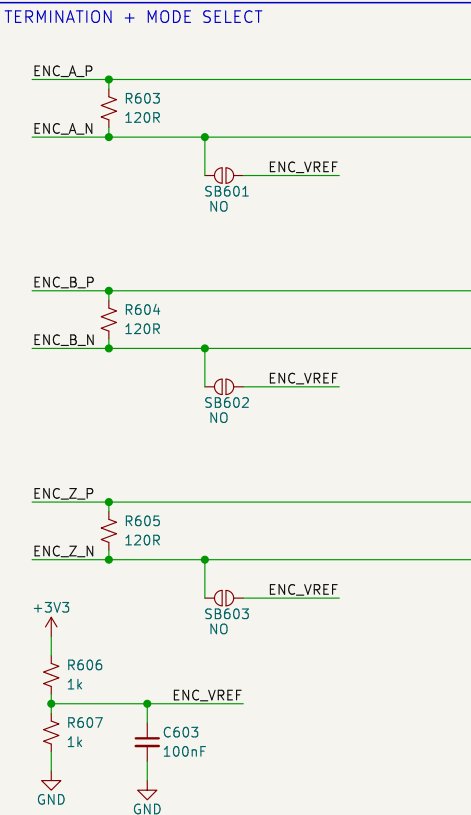
Reasoning + root-sheet needs: docs/decisions/actuator-sch-afe.md		
Requirements: REQ-FF-01..13, REQ-AR-08/09		
HBK S2M 2 mV/V, +/-10 mV FS - one design serves V50N and V10N (DEC-0010)		
ADS1235 gain 128, 4800 SPS Sinc2, 5 V ratiometric bridge excitation		
Amodo Design		
Sheet: /loadcell_afe/		
File: loadcell_afe.kicad_sch		
Title: CBs_1 - Load Cell AFE (ADS1235)		
Size: A3	Date: 2026-09-02	Rev: 0.1
KiCad E.D.A. 9.0.8	Id: 4/10	



J1 – 10-way 0.5 mm FFC, the primary head connection. Mates the stock Bogen cable KABL-FFC-P0.5x10-L100 / -L300.



J2 – 2x5 1.27 mm shrouded header (REQ-PS-09). The same ten signals in parallel with J1, pin n = head pin n. Alternate head connection and the injection point of TEST_PLAN 4.



BLOCK NOTES – linear_encoder

Ordered head: Bogen IKP11-Z1.4-P1-V5-D1-R0.5-F1000-C1 (DEC-0003, REQ-PS-05)
Z1.4 periodic index from the pole pitch, 4 counts P1 1 mm pole pitch
V5 5 V supply, < 65 mA D1 RS-422 A/B/Z
R0.5 0.5 um resolution, post-quadrature F1000 1 MHz per channel
C1 15.8 x 15.4 PCB: a 10-way 0.5 mm FFC connector AND a 1.27 mm pad row

HEAD PIN ORDER (datasheet rev 3.7 p.6 – same order on the FFC and the pads)
1 !PROG 2 SDO 3 /Z 4 Z 5 /B 6 B 7 /A 8 A 9 V+ 10 V-
Check the FFC cable contact side (type A / type D) against J1 before ordering – J1 is a bottom-contact receptacle. J2 needs no such check and is the fallback.

MODES

RS-422, the ordered D1 build (default): R603/R604/R605 fitted, SB601/SB602/SB603 open.
120 R across each pair, as the head datasheet asks (ZO = 120 R at the receiving end). No external fail-safe bias is needed: the AM26LV32 has open-circuit, shorted AND 100 R-terminated fail-safe, so an unplugged or dead head drives a static high – a stalled count, never phantom motion.
Single-ended TTL / bench injection: remove R603/R604/R605, close SB601/SB602/SB603.
Each inverting input then sits on ENC_VREF (1.65 V), so a 0–3.3 V or 0–5 V single-ended source on ENC_xP decodes correctly. This covers a TTL head variant (order code D3) and injection through J2 with no read head fitted.

U1 runs from 3V3, so its outputs are native 3V3 logic (REQ-PS-06); its –0.3 V to +5.5 V input common-mode range accepts the 5 V head directly, and it switches to 32 MHz against the 1 MHz per channel of REQ-PS-07. Both enables are tied active: G (pin 4) high, /G (pin 12) low.

No test points sit on the differential pairs: a probe stub on a terminated 1 MHz RS-422 line is a discontinuity (TEST_PLAN 2.2), and the receiver outputs carry the same information single-ended. J603, the keyed logic-analyser socket, replaces the old TP602.TP607 hooks: CH0..CH4 = LINEAR_ENCODER_A / _B / _Z, ENC_SDO, ENC_nPROG, with GND on 9 and 10 for the ground leads. TP601 stays, a 1.0 mm SMT pad for a DMM on the 5 V head supply.

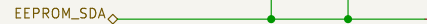
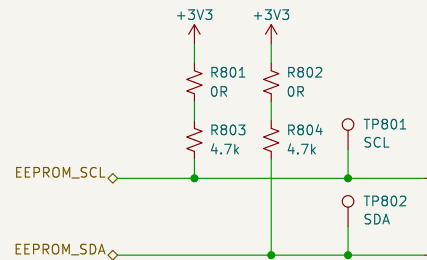
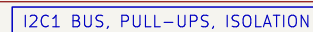
Test strategy: docs/TEST_PLAN.md
REQ-PS-01..10, REQ-AR-07, DEC-0003
RS-422 A/B/Z –> 3V3 into TIM5 (encoder mode TI12 + CH3 capture)
Bogen IKP11-Z1.4-P1-V5-D1-R0.5-F1000-C1 read head interface

Amodo Design

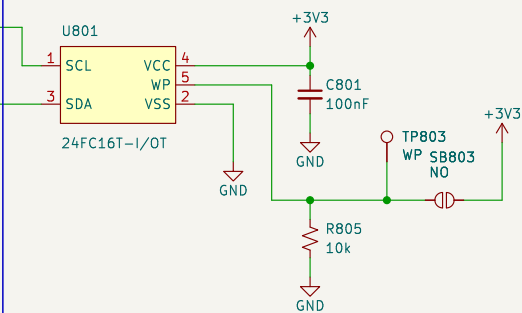
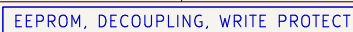
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File: linear_encoder.kicad_sch

Title: CBs_1 – Linear Encoder (IKP11)

Size: A3 Date: 2026-09-02 Rev: 0.2
KiCad E.D.A. 9.0.8 Id: 5/10

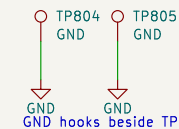


R801/R802 are the removable links of TEST_PLAN 4: lift either to drop its pull-up. SB801/SB802 are cut to take the EEPROM off I2C1 while the bus stays pulled up – this block owns that break, so mcu must not duplicate it. The SDA pull-up crosses the SCL run, no junction.



WP is ACTIVE HIGH. R805 holds it low, so the part is writable and firmware can store calibration at any time; solder SB803 to lock the device permanently. TP3 lets a jig drive WP either way. This resolves OQ-06 – no MCU pin is spent on write protect.

GROUND HOOKS



+3V3 and GND are generated in power_rails and enter this block through the hierarchy; nothing here drives them, so ERC reports power_pin_not_driven until that sheet lands. See docs/decisions/actuator-sch-periph.md.

analyse clips.

BLOCK NOTES – nvm_calibration

U1 = Microchip 24FC16T-I/OT, 16 kbit (2048 x 8) I2C EEPROM in SOT-23-5.

Pin map verified against the datasheet package drawing (DS20001703R p.1):

1 SCL 2 VSS 3 SDA 4 VCC 5 WP.
1.7-5.5 V and 1 MHz capable so it keeps

1.7–5.5 V and 1 MHz capable, so it keeps up with I2C1 at either 400 kHz or Fast-mode Plus, and 4 million erase/write cycles covers repeated calibration.

2 KB is sized for coefficients, not tables: force and temperature calibration constants, the variant (V50N / V10N – see DEC-0010, where the variants differ only by a calibration constant), board serial number and build state. Bulk force profiles live in the OCTOSP11 RAM on the mcu sheet (DEC-0006), not here.

ADDRESSING CAVEAT. The 24xx16 uses all three block-select bits internally, so it answers to the **WHOLE** 1010xxx address space and A0/A1/A2 do not exist on the part. No second 1010-family device can share I2C1. It is the only device on the bus today; if another is ever added, swap to a smaller 24xx02/04 first.

4k7 pull-ups suit 400 kHz on a short board-level bus with one load. They are the only pull-ups on I2C1: the mcu block must not add its own.

Test strategy: docs/TEST_PLAN.md
REQ-EL-08, REQ-AR-11; resolves OQ-06 (write protect)
I2C1: EEPROM_SCL PB8, EEPROM_SDA PB9
16 kbit I2C EEPROM for calibration and compensation data

Amodo Design

Sheet: /nvm_calibration/

File: nvm_calibration.kicad_sch

Title: CBs_1 – NVM / Calibration Store

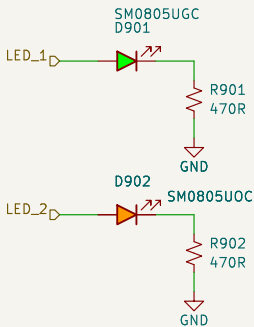
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Rev: 0.2

KiCad E.D.A. 9.0.8

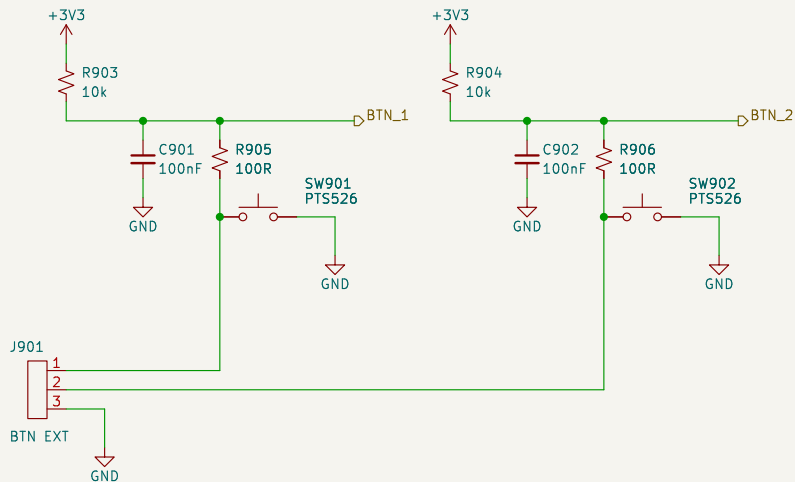
Id: 7/10

STATUS LEDs



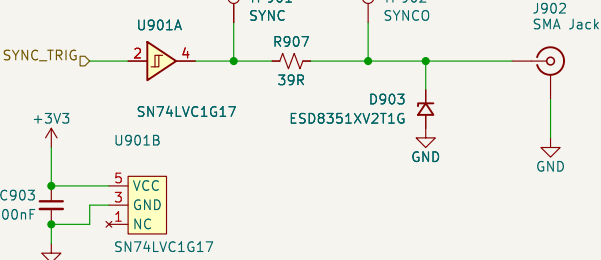
Active high, about 2.8 mA each.
LED_1 green is the heartbeat of
TEST_PLAN bring-up step 3; LED_2
orange is the fault / status lamp.

USER BUTTONS (BTN_1 = HOME)



REQ-CC-06 asks for homing from the API OR an external button, so J901 parallels both on-board switches for a panel button. R905/R906 limit the contact current and set the debounce with C901/C902 – about 1 ms rise, 10 us fall. Which of BTN_1 / BTN_2 is HOME is OQ-05: firmware and silkscreen only, the two channels are electrically identical.

SYNC / TRIGGER OUT (TIM15_CH1)



REQ-EL-05 / REQ-EL-06: 3.3 V push-pull, 50 ohm source terminated, SMA jack. U901 buffers PE5 so the MCU pin never sees the cable and the source impedance is repeatable: R907 39 R plus the LVC output impedance (10–15 ohm typical) is about 50 ohm. TP901 is pre-termination, TP902 post; the SMA itself is the coaxial access of TEST_PLAN 2.3. D903 is 0.55 pF and does not soften the edge.

BLOCK NOTES – ui_io

SAFETY. SPEC 9.1 asks only for a firmware kill on limit actuation (REQ-SF-02). The block diagram and the .ioc both carry a second, independent hardware path (REQ-SF-05, DEC-0012): U902 ANDs the two healthy-limit nodes, so LIMIT_nBRK is HIGH only while both switches are closed and both harness legs are intact. Any limit reached, any broken wire, an unplugged J-LIM or a dead 3V3 rail drives it LOW into TIM1_BKIN2, and the PWM outputs go inactive with firmware taking no part. LIM_A / LIM_B reach the MCU on their own EXTI pins for the firmware kill and for telling the two ends apart. The AND gate's third input is tied high.

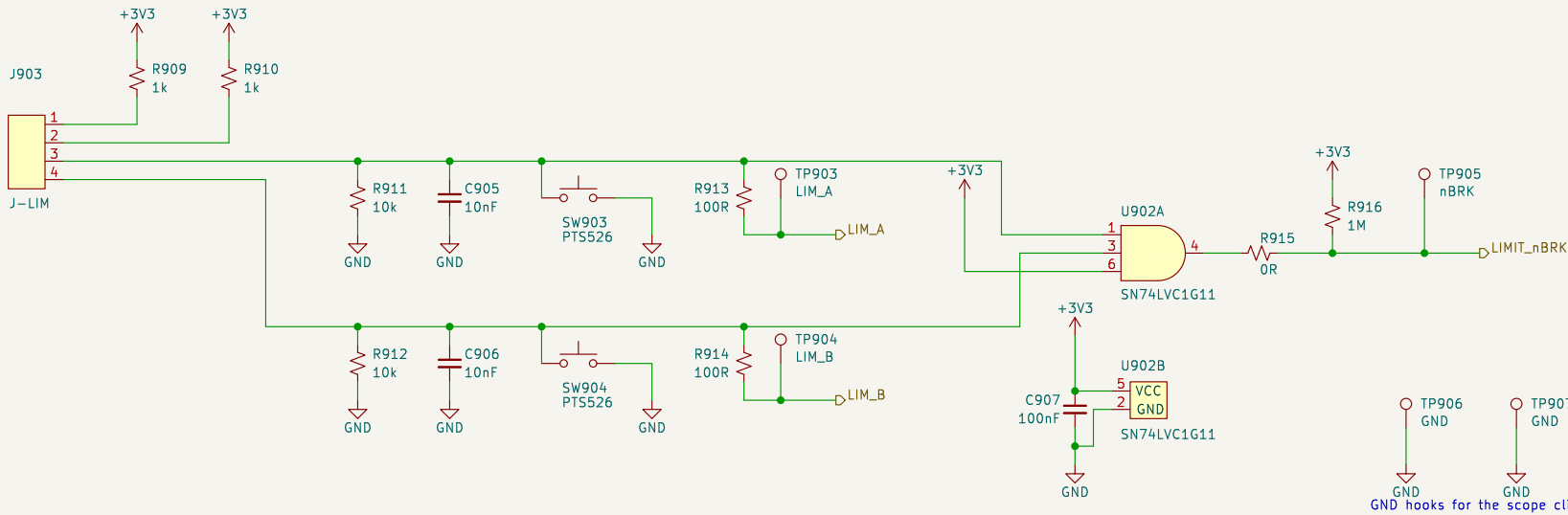
R915 is the TEST_PLAN 3.3 link that lets the two TIM1 BREAK sources be exercised alone. It is fitted by default and must be marked on the silkscreen. With it lifted, R916 (1 M) holds LIMIT_nBRK high so the DRV8323_nFAULT path can be tested on its own; the firmware kill through LIM_A / LIM_B is unaffected either way. 1 M is weak enough that a tripped chain still pulls the net down to about 30 mV.

SW903 / SW904 are the 'assert each limit without the mechanics' means of TEST_PLAN 4: pressing one pulls its node low through the 1 k feed (3.3 mA), which is exactly the state that end of stroke produces. They act independently because the two switch loops are independent – a series safety chain could not be stimulated one end at a time.

C905 / C906 (10 nF) with the -0.9 k source give about 9 us of rise filtering and 100 us of fall; at the 20 mm/s of REQ-ME-03 that is 2 um of extra travel before the break asserts, against millimetres of switch overtravel. R913 / R914 protect the MCU pins. No discrete TVS is fitted on the limit lines: the harness stays inside the instrument and the RC plus the series resistor already bound the current into the MCU clamps. The SMA is the one externally exposed pin here, and it does get a clamp (D903).

+3V3 and GND are generated in power_rails and enter this block through the hierarchy; nothing here drives them, and power_rails owns their PWR_FLAGS, so project ERC is clean at severity-all.

END-OF-STROKE LIMIT SWITCHES + TIM1 BREAK



J-LIM pinout: 1 LIM_A feed 2 LIM_B feed 3 LIM_A return 4 LIM_B return.
Both switches are NORMALLY CLOSED, so a closed switch pulls its node HIGH (3.0 V) and an open switch – or a broken wire, or an unplugged connector – lets the 10 k pull-down take it LOW. LOW therefore means 'at that end of stroke, or the wiring has failed', which is the safe direction to fail in.

Test strategy: docs/TEST_PLAN.md
REQ-EL-05/06/09, REQ-CC-06, REQ-SF-01/02/05
LIMIT_nBRK PE6 -> TIM1_BKIN2: hardware motor kill (DEC-0012)
Status LEDs, user buttons, SMA SYNC output, end-of-stroke limits

Amodo Design

Sheet: /ui_io/
File: ui_io.kicad_sch

Title: CBs_1 – User I/O, SYNC, Limit Switches

Size: A3 Date: 2026-09-02 Rev: 0.2
KiCad E.D.A. 9.0.8 Id: 8/10

Root sheet wired by the integration pass (DEC-0009)
 Requirements: docs/REQUIREMENTS.md Tests: docs/TEST_PLAN.md
 Reasoning: docs/decisions/actuator-sch-motor.md
 3-phase inverter, phase-current sense, motor and rotary-encoder interface

Sheet: /motor_drive/
File: motor_drive.kicad_sch

Size: A3	Date: 2026-09-03	Rev: 0.2
KiCad E.D.A. 9.0.8		Id: 10/10

Size: A3

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Date: 2026-09-03

Rev: 0.2

Id: 10/10